

Network Security

Cyber Security

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Network Security

Network Security refers to the protection of networks from data which they carry from unauthorized access which may jeopardize the network. It also protects from misuse, and cyberattacks including, malware, ransomware, phishing, Distributed Denial of Service and Man-in-the-Middle attacks and many more. This is achieved through a combination of policies, technologies, people and procedures that work together to defend any communication infrastructure from cyberattacks. This layered approach ensures data integrity and reliable performance across increasingly complex, distributed networks. All of this is done through upholding the benefits of the CIA Triad which includes, Confidentiality, Integrity and Availability.

Network security does not only refer to the protection of the network infrastructure itself but it also means securing traffic and network-accessible assets at both the network-edge and inside the perimeter. It works through multiple layers that control the network access at the network edge and inside the environment. The layers are secured to ensure that at each layer only authorized users can access the resources whilst simultaneously blocking unauthorized access which may be a threat to the network. These layers include the Physical Network Security, Administrative Network Security and Technical Network Security.

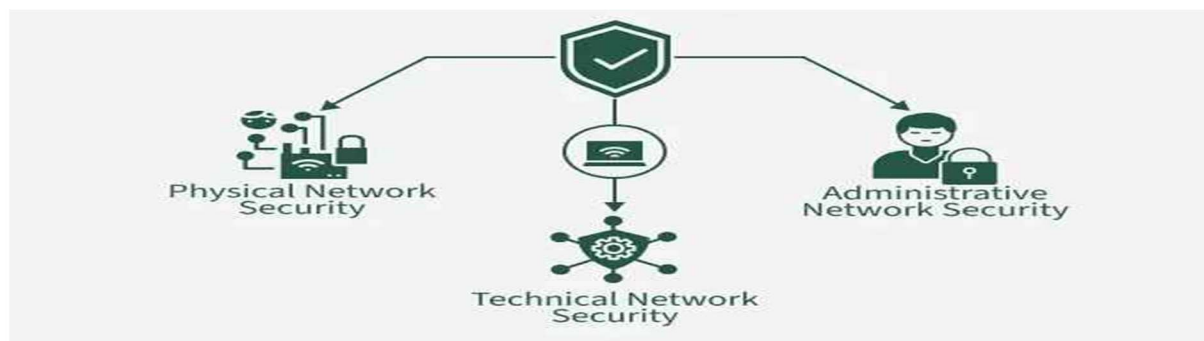


Figure 1: Network Security Layers

The network threats discussed above can happen very easily and often have devastating effects to the those affected. These threats lead to financial loss, theft of

identity, data breaches and system downtime. To be able to defend against these network threats, one needs to understand them and how they differ from each other. Understanding network threats is crucial for organizations and individuals to come up with mitigation strategies and security measures such as strong passwords, firewalls, encryption and traffic monitoring.

Types of Network Threats

Viruses

A virus is a type of malicious software that attaches itself to legitimate files or systems and aims to disrupt systems, corrupt files, slow down performance, information theft and cause data losses and leakages (Fortinet , 2026). It is intentionally made to appear as a normal file, for example, an email one receives with an executable host file attachment which upon opening the file, the user exposes their computer to executable malicious code which then spreads to important files and systems within the computer and cause serious damages. Viruses can also be spread through infected USB devices, malicious websites and software downloads containing executable malicious code (Malwarebytes , 2023). They can spread from one device to another when unsuspecting users interact with infected devices, much like the human flu virus (Corrons, 2024).

Although computer viruses are not too harmful, they can still significantly slow down the performance of your computer as they use up a lot of memory and may lead to the computer crashing more often (Geeksforgeeks , 2025). Some cases are a bit more severe with some viruses creating more copies of themselves and sending so much stuff over the internet to the point where it becomes almost impossible for the user to access the internet or do anything online (Geeksforgeeks , 2025).

Worms

A subset of the Trojan Horse malware as it self-replicates from one computer to another without needing human intervention (Malwarebytes , 2026). The computer worm originated as an experiment gone wrong and has since ascended to a dangerous tool in the hacker space and for cybersecurity professionals (Birchall, 2025). Unlike Viruses, computer worms are standalone programs that do not require attachment or to be disguised or hidden into other programs and files to cause damage (Birchall, 2025). This makes them extremely dangerous as they spread fast. A computer worm operates by first finding a vulnerability in a network, be it through software backdoors, file transfers, email attachments, malicious links hidden in pop-up ads or other network vulnerabilities (Birchall, 2025). Once installed, they self-replicate, spreading to files and systems on the computer and even going as far as spreading to other devices within the network.

Worms consume system resources and computer bandwidth, leading to performance issues and system errors (Malwarebytes , 2026). Severe damages can also be caused by computer worms including, corrupting, modifying or deleting files and opening backdoors that allow hackers to steal sensitive data (Birchall, 2025). Things like keeping your software updated and using strong passwords, keeping an enabled or configured firewall on your devices, not clicking on links in suspicious emails, avoiding clicking pop-up adds or suspicious advertisements and installing an antivirus and anti-malware software can help prevent computer worm infection (Geeksforgeeks , 2025).

Trojans

Unlike viruses and worms, Trojans do not self-replicate but instead rely on deception. Trojans masquerade as legitimate software to trick users into downloading malicious software, typically spread through social engineering tactics such as phishing, their payload usually consists of a backdoor that provides the attacker with unauthorised access to the infected computer, potentially granting access to personal information (TitanFile, 2025). The U.S. Cybersecurity and

Infrastructure Security Agency confirmed that in 2021 the top malware strains included remote access Trojans (RATs), banking Trojans, information stealers, and ransomware (CISA, 2022). Banking Trojans in particular are notably sophisticated, for example, Ursnif is a banking Trojan that steals financial information and has evolved over the years to include a persistence mechanism, methods to avoid sandboxes and virtual machines, and search capability for disk encryption software, typically delivered as a malicious attachment to phishing emails (CISA, 2022).

Phishing attacks

Phishing represents one of the most economically destructive and widespread threats in contemporary cybersecurity. Since the first reported phishing attack in 1990, it has evolved into a more sophisticated attack vector, and at present is considered one of the most frequent examples of fraud activity on the Internet. Phishing attacks are the most common type of cybersecurity breach as stated by official statistics from the Cybersecurity Breaches Survey 2020 in the United Kingdom (Alkhalil et al., 2021).

Technically, phishing is an identity theft mechanism that deceives internet users into revealing their sensitive data, such as login information and credit or debit card details (Jalali et al., 2021). The financial consequences are severe, phishing appeared as the costliest attack vector in 2022 with an average cost of US \$4.91 million per data breach, and organisations suffer in terms of the cost to contain malware, productivity losses, the cost of credential compromise, and the cost of ransomware delivered via phishing, in addition to reputational damage (Alahmadi et al., 2023).

The attack vector continues to evolve in sophistication. Organisations should expect phishing to remain one of the main threat vectors that hackers use to deliver both ransomware and business email compromise (BEC) attacks, with cybercriminals employing increasingly subtle techniques such as embedding malicious content within "unsubscribe" captions in emails to confirm the validity of an address and lead victims to further spam (Carroll, Adejobi and

Montasari, 2022). The COVID-19 pandemic accelerated this trend considerably, as the shift to remote working and online activity created significant new opportunities for phishing campaigns (Carroll, Adejobi and Montasari, 2022).

Security Concepts

Firewalls

Firewalls are the foundational control mechanism of network defence. In network security, a firewall is a security system that observes and controls network traffic based on predefined rules, setting up a barrier between an internal network and an outside unsecured network such as the Internet (Zia and Ali, 2018). Modern deployments increasingly favour next-generation firewalls (NGFWs), an NGFW will identify potential inbound threats and outgoing data exfiltration and block malicious data flows from crossing the network boundary, making it the cornerstone of any network security architecture (Check Point Software, 2025).

Firewalls should be deployed alongside network segmentation. The U.S. National Security Agency recommends that similar systems be isolated into different subnets or virtual local area networks (VLANs), or physically separated via firewalls or filtering routers, with workstations, servers, printers, telecommunication systems, and other network peripherals kept separate from each other (NSA, 2023). Research into the integration of firewalls with other controls has found that while encryption secures data from being understood by unauthorised users, it does not provide security against denial-of-service attacks or packet dropping, firewalls protect data in such cases, though the integration of both mechanisms introduces performance trade-offs that must be managed carefully (International Journal of Scientific Research in Computer Science, Engineering and Information Technology, 2022).

Encryption

Encryption is central to protecting data both in transit and at rest. SSL (Secure Sockets Layer) and its more secure and updated successor, TLS (Transport Layer Security), are encryption and authentication protocols designed for secure communication over computer networks (IEEE, 2023). TLS has become the dominant standard. TLS has become the mainstream encryption protocol used to secure communications for every HTTP site, and with the release of TLS 1.3 and the adoption of DNS-over-HTTPS and Encrypted SNI, nearly 90% of internet traffic is now encrypted (Guan, Li and Xiong, 2022).

However, the near universal adoption of encryption introduces new security challenges. Security operations centres (SOCs) face a critical issue in that the increased adoption of TLS encryption on the internet significantly complicates network traffic analysis, as malicious traffic can be concealed within encrypted payloads, with deployment scenarios for malware detection and family classification requiring increasingly sophisticated analytical approaches (Lansky et al., 2021). Empirically, the protective value of encryption in organisational settings has been confirmed. A study of IT professionals overseeing e-commerce operations found that encryption strength, firewall configuration, and authentication protocols are significant cybersecurity measures in protecting platforms against cyber-attacks (Al Naim and Ghouri, 2023).

Secure Network Configuration

Secure network configuration applies the principle of defence in depth, layering multiple controls so that no single failure compromises the entire system. The concept of defence in depth suggests multiple security layers ensuring additional protection even if one layer fails. This is complemented by continuous monitoring, which is vital to identify potential vulnerabilities and respond to evolving cyber threats proactively (Nwakeze, 2024). A comprehensive set of measures is required, organisations can implement firewalls, intrusion detection systems (IDS), encryption, access control, VPNs, and security auditing and monitoring

to secure their network infrastructure and maintain the integrity of sensitive information (Ahmad, 2023).

Patch management is an indispensable configuration discipline. Many threats, including worms, exploit unpatched vulnerabilities to spread to new systems, and promptly applying updates and patches helps to close security gaps before an attacker can exploit them (Check Point Software, 2025). Beyond patching, zero-trust architecture represents an important evolution in secure configuration thinking. The zero-trust network approach, with identity authentication as its core mechanism for access control, solves the security problems of traditional IP networks where passive, blocking-and-killing defensive measures cannot fundamentally meet modern network security requirements (Towards Double Defense Network Security, 2022).

Software and Tools used for this Practical.

- Wireshark was used in this practical to capture and analyse network traffic.
- Windows Defender Firewall was utilized to configure and manage firewall security rules.
- To view network configuration and information, Command Prompt was used.

Viewing Network Configuration

Network configuration is an important procedure in the cybersecurity space. It establishes network situational awareness as it analyses the host's logical identity and routing boundaries within the network architecture. This information is especially useful during investigations, penetration testing and traffic monitoring.

The command `ipconfig` was executed in Command Prompt and returned the IPv4 address, subnet mask, default gateway, and network adaptor information. The IPv4 address represents the device on the network while the default gateway identifies the gateway used to access the internet.

```

Wireless LAN adapter Wi-Fi:

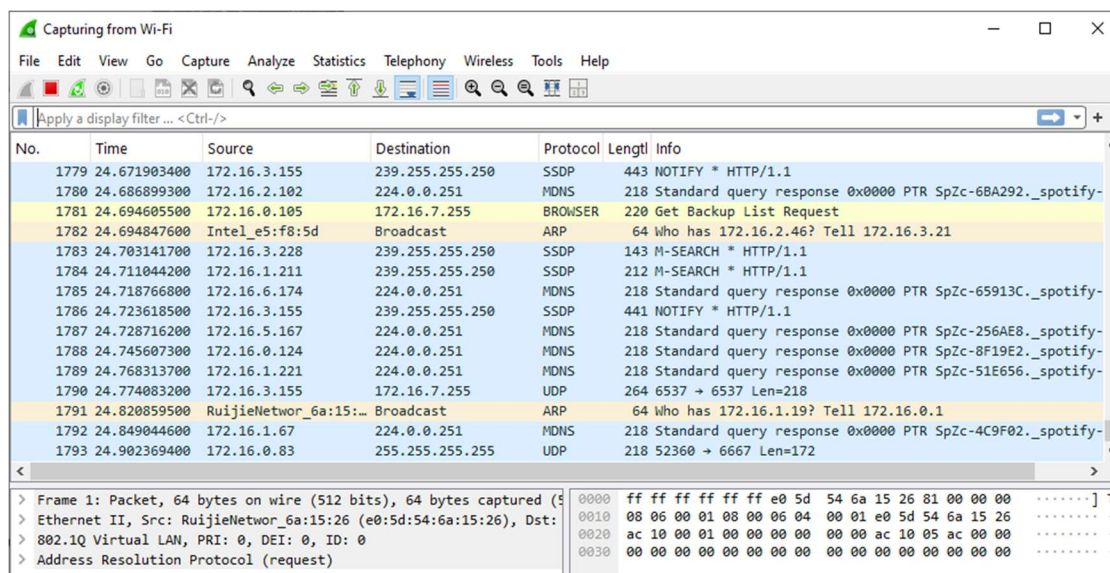
Connection-specific DNS Suffix  . : lan
IPv6 Address. . . . . : fd08:bd40:b663:41cf:9e10:4ffa:c2fd:20b4
Temporary IPv6 Address. . . . . : fd08:bd40:b663:41cf:ded:8585:d51c:f91e
Link-local IPv6 Address . . . . . : fe80::84b9:6a19:8ee4:2750%11
IPv4 Address. . . . . : 172.16.4.213
Subnet Mask . . . . . : 255.255.248.0
Default Gateway . . . . . : 172.16.0.1

```

The active IPv4 address and the Default gateway handling external web traffic are managed under the wireless LAN adaptor profile since the host system connects to the target network using a wireless interface.

Capturing Network Traffic using Wireshark

Wireshark was used to capture packets in real time. The active network interface was selected so that packets are captured in real time. To achieve this, several websites were visited in order to generated network traffic.



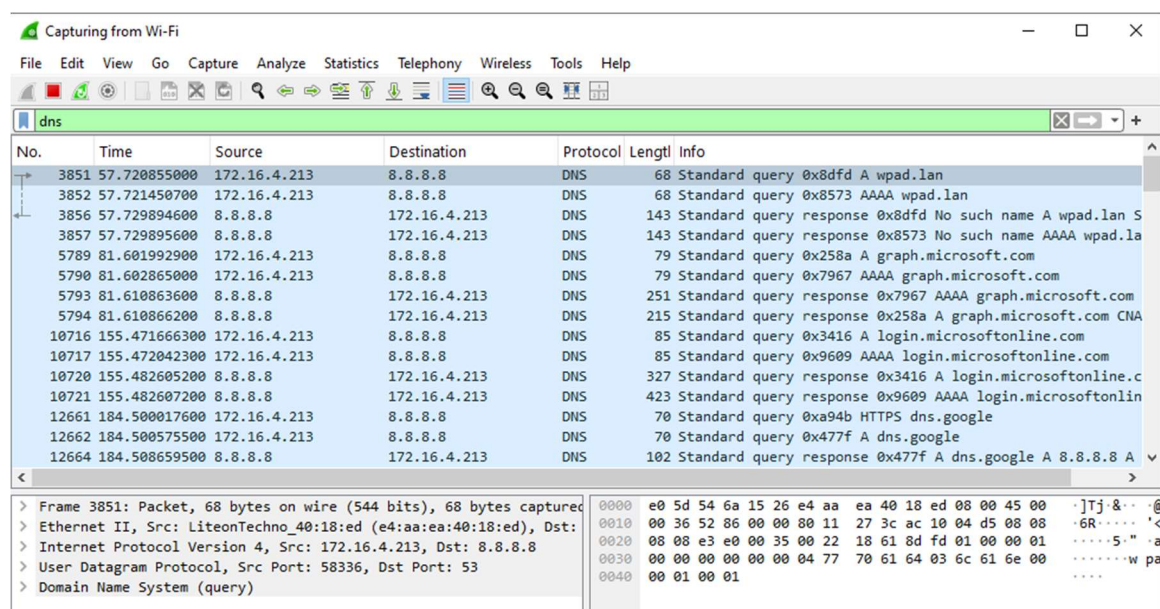
Packet analyses is widely used In cybersecurity as it allows analysts to monitor communication to identify suspicious behaviour, recognize malicious communication, fix network issues and for investigation of cyber incidents.

The screenshot above shows Wireshark open, and captures real-time broadcasting of network traffic moving across a wireless network segment. Local devices on the network

such as the ones running Spotify are seen broadcasting their presence or to discover smart devices on the same subnet.

The packet details pane on the bottom shows a 802.1Q Virtual LAN header, proving that the network traffic is running on a configured virtual LAN segmentation layer.

DNS Traffic



No.	Time	Source	Destination	Protocol	Length	Info
3851	57.720855000	172.16.4.213	8.8.8.8	DNS	68	Standard query 0x8dfd A wpad.lan
3852	57.721450700	172.16.4.213	8.8.8.8	DNS	68	Standard query 0x8573 AAAA wpad.lan
3856	57.729894600	8.8.8.8	172.16.4.213	DNS	143	Standard query response 0x8dfd No such name A wpad.lan S
3857	57.729895600	8.8.8.8	172.16.4.213	DNS	143	Standard query response 0x8573 No such name AAAA wpad.la
5789	81.601992900	172.16.4.213	8.8.8.8	DNS	79	Standard query 0x258a A graph.microsoft.com
5790	81.602865000	172.16.4.213	8.8.8.8	DNS	79	Standard query 0x7967 AAAA graph.microsoft.com
5793	81.610863600	8.8.8.8	172.16.4.213	DNS	251	Standard query response 0x7967 AAAA graph.microsoft.com
5794	81.610866200	8.8.8.8	172.16.4.213	DNS	215	Standard query response 0x258a A graph.microsoft.com CNA
10716	155.471666300	172.16.4.213	8.8.8.8	DNS	85	Standard query 0x3416 A login.microsoftonline.com
10717	155.472042300	172.16.4.213	8.8.8.8	DNS	85	Standard query 0x9609 AAAA login.microsoftonline.com
10720	155.482605200	8.8.8.8	172.16.4.213	DNS	327	Standard query response 0x3416 A login.microsoftonline.c
10721	155.482607200	8.8.8.8	172.16.4.213	DNS	423	Standard query response 0x9609 AAAA login.microsoftonlin
12661	184.500017600	172.16.4.213	8.8.8.8	DNS	70	Standard query 0xa94b HTTPS dns.google
12662	184.500575500	172.16.4.213	8.8.8.8	DNS	70	Standard query 0x477f A dns.google
12664	184.508659500	8.8.8.8	172.16.4.213	DNS	102	Standard query response 0x477f A dns.google A 8.8.8.8 A

> Frame 3851: Packet, 68 bytes on wire (544 bits), 68 bytes captured
> Ethernet II, Src: LiteonTechno_40:18:ed (e4:aa:ea:40:18:ed), Dst:
> Internet Protocol Version 4, Src: 172.16.4.213, Dst: 8.8.8.8
> User Datagram Protocol, Src Port: 58336, Dst Port: 53
> Domain Name System (query)

The Domain Name System (DNS) acts as the phonebook of the internet by taking human readable text and converting it into numerical IP destinations that machines require to route traffic. For example, Google's "8.8.8.8" as seen in the picture above.

Analysing the DNS is crucial because attackers sometimes abuse DNS communication for malware communication, phishing attacks, DNS tunnelling and spoofing. Analysing DNS traffic helps identify suspicious domains and communications that might be malicious. The traffic was filtered to only show DNS packets, as displayed above. Website queries are also visible as well as source and destination addresses.

HTTP Traffic

No.	Time	Source	Destination	Protocol	Length	Info
22850	-692.7225889...	172.16.4.213	5.188.133.99	HTTP	336	GET /msdownload/update/v3/static/trustedr/en/pinrulesstl.c...
22852	-692.7138866...	5.188.133.99	172.16.4.213	HTTP	446	HTTP/1.1 304 Not Modified
27996	-619.6949378...	fe80::8cb:8990:8325...	fe80::84b9:6a19:8ee...	HTTP/X...	807	POST /f5bccdd6e-9abe-4f34-aa2b-c8b41ba4a396/ HTTP/1.1
27999	-619.6912785...	fe80::84b9:6a19:8ee...	fe80::8cb:8990:8325...	HTTP/X...	985	HTTP/1.1 200

Frame 22850: Packet, 336 bytes on wire (2688 bits), 336 bytes captured (2688 bits) on interface 0
Ethernet II, Src: LiteonTechno_40:18:ed (e4:aa:ea:40:18:ed), Dst: 5.188.133.99
Internet Protocol Version 4, Src: 172.16.4.213, Dst: 5.188.133.99
Transmission Control Protocol, Src Port: 65242, Dst Port: 80, Seq: 336, Win: 0, Len: 0
Hypertext Transfer Protocol

HTTP stands for Hypertext Transfer Protocol. It is a system that allows communication between web browsers like Google or Firefox as well as web servers. It is a set of protocols or rules that govern the communication of user browser and web server communication ensuring that websites load correctly.

This filter demonstrated communication between web browsers and web servers. HTTP traffic appears to be limited because modern websites use HTTPS encryption. The modern use of HTTPS is because of its encryption unlike HTTP which is not encrypted. Filtering traffic to analyse http traffic is especially important as HTTP traffic may expose sensitive information such as usernames and passwords. HTTP analyses helps analysts identify insecure traffic and detect suspicious requests.

TCP Traffic

No.	Time	Source	Destination	Protocol	Length	Info
22562	307.776016100	172.16.3.155	172.16.4.213	TLSv1.2	164	Application Data
22573	307.835627300	172.16.4.213	172.16.3.155	TCP	54	62251 → 8009 [ACK] Seq=6821 Ack=6821 Win=511 Len=0
22616	308.668364500	172.16.4.213	74.242.255.116	TCP	55	[TCP Keep-Alive] 65057 → 443 [ACK] Seq=67 Ack=60 Win=516
22626	308.849671900	74.242.255.116	172.16.4.213	TCP	66	[TCP Keep-Alive ACK] 443 → 65057 [ACK] Seq=60 Ack=68 Win=
22843	311.163802700	172.16.4.213	5.188.133.99	TCP	66	65242 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 S
22848	311.170392800	5.188.133.99	172.16.4.213	TCP	66	80 → 65242 [SYN, ACK] Seq=0 Ack=1 Win=32120 Len=0 MSS=14
22849	311.170560000	172.16.4.213	5.188.133.99	TCP	54	65242 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0
22850	311.170801800	172.16.4.213	5.188.133.99	HTTP	336	GET /msdownload/update/v3/static/trusted/en/pinrulesstl
22851	311.179367200	5.188.133.99	172.16.4.213	TCP	60	80 → 65242 [ACK] Seq=1 Ack=283 Win=32768 Len=0
22852	311.179504100	5.188.133.99	172.16.4.213	HTTP	446	HTTP/1.1 304 Not Modified
22853	311.235945600	172.16.4.213	5.188.133.99	TCP	54	65242 → 80 [ACK] Seq=283 Ack=393 Win=130816 Len=0
22903	312.024321400	172.16.4.213	150.171.28.12	TCP	54	65241 → 443 [RST, ACK] Seq=3568 Ack=30833 Win=0 Len=0
22972	312.781767500	172.16.4.213	172.16.3.155	TLSv1.2	164	Application Data
22973	312.795725100	172.16.3.155	172.16.4.213	TLSv1.2	164	Application Data
22976	312.846878500	172.16.4.213	172.16.3.155	TCP	54	62251 → 8009 [ACK] Seq=6931 Ack=6931 Win=511 Len=0

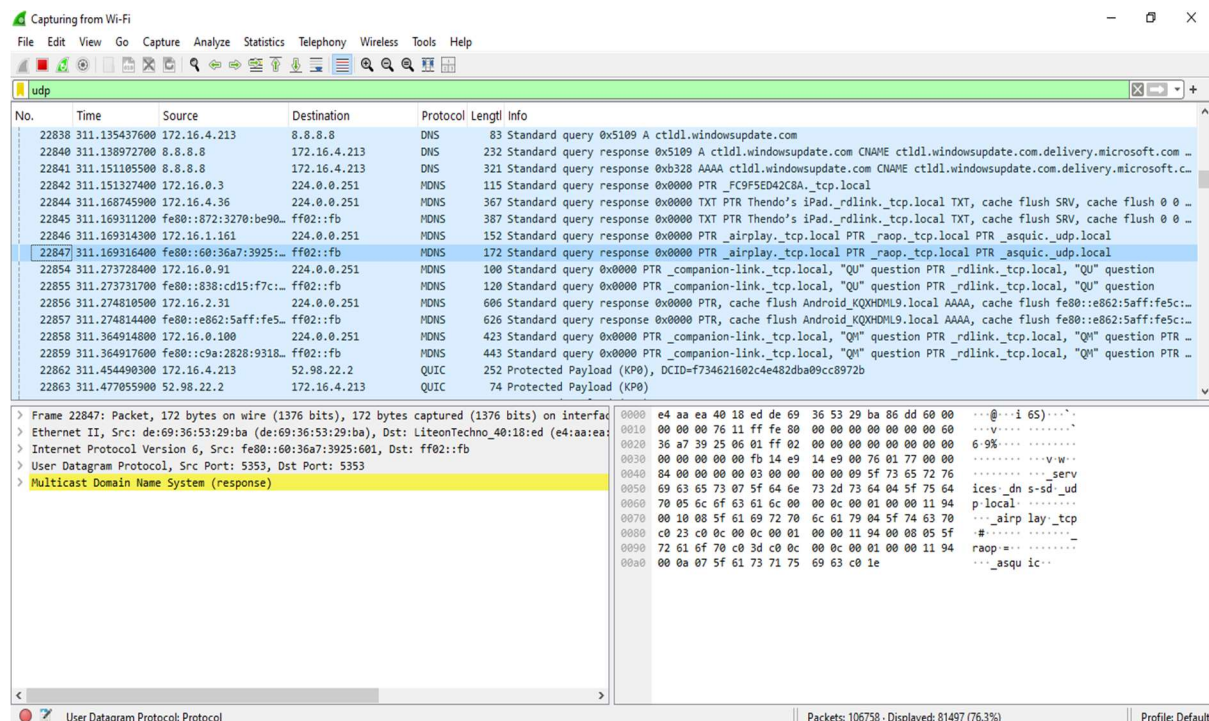
Frame 22850: Packet, 336 bytes on wire (2688 bits), 336 bytes captured on interface 0
Ethernet II, Src: LiteonTechno_40:18:ed (e4:aa:ea:40:18:ed), Dst: 5.188.133.99
Internet Protocol Version 4, Src: 172.16.4.213, Dst: 5.188.133.99
Transmission Control Protocol, Src Port: 65242, Dst Port: 80, Seq: 65242, Win: 64240, Len: 0
Hypertext Transfer Protocol

Transmission Control Protocol (TCP) is a protocol that allows devices to communicate reliably over a network. It requires a structured three-way-handshake exchanging SYN, SYN-ACK, and ACK flags to establish a connection before any actual data can be sent, ensuring that both the sender and the receiver are ready.

This lifecycle can be observed from the above picture from packets 22843, 22848, and 22849 as they capture my host machine (172.16.4.213) executing a handshake with an external server (5.188.133.99) over port 80.

The above traffic demonstrates reliable communication between systems. TCP analysis is crucial as many cyberattacks use TCP communication including scanning attacks, unauthorised connection attempts, malicious communication and denial-of-service attacks

UDP Traffic



User Datagram Protocol (UDP) is a transport layer protocol of the Internet Protocol (IP) that provides fast, connectionless, and lightweight communication between processes. Delivery, order, or error checking are not guaranteed, making it a very suitable protocol for real-time and time-sensitive applications such as video streaming and online gaming.

The applied UDP filter has highlighted a massive stream of local Multicast DNS traffic targeting the Multicast destination IP 224.0.0.251 over UDP port 5353, where personal devices such as Apple AirPlay services and Android devices are visible in the packet info constantly shouting their presence to the local network.

Identifying Suspicious Traffic

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn == 1 && tcp.flags.ack == 0

No.	Time	Source	Destination	Protocol	Length	Info
89620	1269.7518489..	172.16.2.236	172.16.4.213	TCP	66	55339 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
89665	1270.7611641..	172.16.2.236	172.16.4.213	TCP	66	[TCP Retransmission] 55339 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
89744	1271.7977338..	172.16.4.213	23.196.227.17	TCP	66	65271 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
89859	1272.7834590..	172.16.2.236	172.16.4.213	TCP	66	[TCP Retransmission] 55339 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
90208	1276.7876704..	172.16.2.236	172.16.4.213	TCP	66	[TCP Retransmission] 55339 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
90782	1284.7920040..	172.16.2.236	172.16.4.213	TCP	66	[TCP Retransmission] 55339 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
91736	1298.9322668..	172.16.4.213	135.233.95.80	TCP	66	65272 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
91918	1300.6796633..	172.16.4.213	150.171.22.17	TCP	66	65273 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
102398	1451.1690003..	172.16.4.213	52.178.17.3	TCP	66	65274 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
117420	1656.6462741..	172.16.4.213	20.190.190.35	TCP	66	65275 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
117421	1656.6464296..	172.16.4.213	20.190.190.35	TCP	66	65276 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
117758	1661.7285752..	172.16.4.213	140.82.114.25	TCP	66	65277 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
117923	1664.6142312..	172.16.4.213	20.87.245.0	TCP	66	65278 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
117969	1664.9628336..	172.16.4.213	140.82.114.25	TCP	66	65279 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
124768	1757.9443906..	172.16.3.88	172.16.4.213	TCP	66	50324 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
124800	1758.6297567..	172.16.3.88	172.16.4.213	TCP	66	50326 → 445 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

> Frame 527: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0x...
> Ethernet II, Src: LiteonTechno_40:18:ed (e4:aa:ea:40:18:ed), Dst: RuijieNetwork_6a:15:26 (e0:54:00:12:35:52)
> Internet Protocol Version 4, Src: 172.16.4.213, Dst: 23.196.227.9
> Transmission Control Protocol, Src Port: 65232, Dst Port: 443, Seq: 0, Len: 0

0000 e0 5d 54 6a 15 26 e4 aa ea 40 18 ed 00 00 45 00 ..TJ&...@...E
0010 00 34 4e f5 40 00 00 06 00 1c ac 10 04 d5 17 c4 -4N@...
0020 e3 09 fe d0 01 bb 32 97 8c b0 00 00 00 00 02 ..-2-.....
0030 fa f0 08 99 00 00 02 04 05 b4 01 03 03 00 01 01
0040 04 02 ..

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn == 1 && tcp.flags.ack == 0

No.	Time	Source	Destination	Protocol	Length	Info
124800	1758.6297567..	172.16.3.88	172.16.4.213	TCP	66	50326 → 445 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
124803	1758.6297630..	172.16.3.88	172.16.4.213	TCP	66	50325 → 62078 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
124814	1758.5287545..	172.16.3.88	172.16.4.213	TCP	66	[TCP Retransmission] 50324 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
124888	1759.6444086..	172.16.3.88	172.16.4.213	TCP	66	[TCP Retransmission] 50326 → 445 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
124889	1759.6444220..	172.16.3.88	172.16.4.213	TCP	66	[TCP Retransmission] 50325 → 62078 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125016	1761.6451344..	172.16.3.88	172.16.4.213	TCP	66	[TCP Retransmission] 50325 → 62078 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125017	1761.6497115..	172.16.3.88	172.16.4.213	TCP	66	[TCP Retransmission] 50326 → 445 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125052	1762.1241232..	172.16.3.88	172.16.4.213	TCP	66	50335 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125062	1762.2743063..	172.16.3.88	172.16.4.213	TCP	66	50336 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125087	1762.5571037..	172.16.3.88	172.16.4.213	TCP	66	64261 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125097	1762.6544687..	172.16.3.88	172.16.4.213	TCP	66	65322 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125100	1762.7259903..	172.16.3.88	172.16.4.213	TCP	66	65323 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125116	1762.8673225..	172.16.3.88	172.16.4.213	TCP	66	65324 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125135	1763.0072667..	172.16.3.88	172.16.4.213	TCP	66	65325 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125146	1763.1051865..	172.16.3.88	172.16.4.213	TCP	66	65095 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
138781	1969.2367690..	172.16.4.213	13.69.239.72	TCP	66	65280 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

> Frame 527: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0x...
> Ethernet II, Src: LiteonTechno_40:18:ed (e4:aa:ea:40:18:ed), Dst: RuijieNetwork_6a:15:26 (e0:54:00:12:35:52)
> Internet Protocol Version 4, Src: 172.16.4.213, Dst: 23.196.227.9
> Transmission Control Protocol, Src Port: 65232, Dst Port: 443, Seq: 0, Len: 0

0000 e0 5d 54 6a 15 26 e4 aa ea 40 18 ed 00 00 45 00 ..TJ&...@...E
0010 00 34 4e f5 40 00 00 06 00 1c ac 10 04 d5 17 c4 -4N@...
0020 e3 09 fe d0 01 bb 32 97 8c b0 00 00 00 00 02 ..-2-.....
0030 fa f0 08 99 00 00 02 04 05 b4 01 03 03 00 01 01
0040 04 02 ..

Suspicious traffic was recognized around the 1757 second mark. The internal host suddenly targets 172.16.4.213 with an astounding succession of SYN packets across different service ports in the span of milliseconds. It hits port 80, port 445, and port 62078 all at once. Immediately after that, it shifts and begins rapidly sending SYN requests to port 5357.

Capturing from Wi-Fi

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tcp.flags.syn == 1 && tcp.flags.ack == 0

No.	Time	Source	Destination	Protocol	Length	Info
153792	2207.1783324..	172.16.4.213	23.196.227.13	TCP	66	65294 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
153946	2209.4308091..	172.16.4.213	5.188.133.99	TCP	66	65295 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
154160	2213.5525121..	172.16.4.213	5.188.133.99	TCP	66	65296 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
154957	2224.6957278..	172.16.4.213	52.98.20.178	TCP	66	65297 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
156142	2239.6660614..	172.16.4.213	5.188.133.99	TCP	66	65298 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
162638	2344.7450856..	172.16.4.213	52.98.18.34	TCP	66	65299 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
163565	2358.6353149..	172.16.4.213	13.89.179.11	TCP	66	65300 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170248	2476.7125440..	172.16.1.145	172.16.4.213	TCP	66	55142 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170328	2477.7178586..	172.16.1.145	172.16.4.213	TCP	66	[TCP Retransmission] 55142 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170390	2478.9113676..	172.16.1.145	172.16.4.213	TCP	66	55148 → 62078 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170391	2478.9113706..	172.16.1.145	172.16.4.213	TCP	66	55149 → 445 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170405	2479.1277153..	172.16.1.145	172.16.4.213	TCP	66	55150 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170413	2479.2161211..	172.16.1.145	172.16.4.213	TCP	66	55151 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170422	2479.4258692..	172.16.1.145	172.16.4.213	TCP	66	55152 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170450	2479.8140117..	172.16.1.145	172.16.4.213	TCP	66	55153 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170455	2479.9164007..	172.16.1.145	172.16.4.213	TCP	66	[TCP Retransmission] 55149 → 445 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

> Frame 527: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0x...
> Ethernet II, Src: LiteonTechno_40:18:ed (e4:aa:ea:40:18:ed), Dst: RuijieNetwork_6a:15:26 (e0:54:00:12:35:52)
> Internet Protocol Version 4, Src: 172.16.4.213, Dst: 23.196.227.9
> Transmission Control Protocol, Src Port: 65232, Dst Port: 443, Seq: 0, Len: 0

0000 e0 5d 54 6a 15 26 e4 aa ea 40 18 ed 00 00 45 00 ..TJ&...@...E
0010 00 34 4e f5 40 00 00 06 00 1c ac 10 04 d5 17 c4 -4N@...
0020 e3 09 fe d0 01 bb 32 97 8c b0 00 00 00 00 02 ..-2-.....
0030 fa f0 08 99 00 00 02 04 05 b4 01 03 03 00 01 01
0040 04 02 ..

Capturing from Wi-Fi

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tcp.flags.syn == 1 && tcp.flags.ack == 0

No.	Time	Source	Destination	Protocol	Length	Info
170455	2479.9164007..	172.16.1.145	172.16.4.213	TCP	66	[TCP Retransmission] 55149 → 445 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170456	2479.9164031..	172.16.1.145	172.16.4.213	TCP	66	[TCP Retransmission] 55148 → 62078 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170621	2481.9240164..	172.16.1.145	172.16.4.213	TCP	66	[TCP Retransmission] 55148 → 62078 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170623	2481.9291096..	172.16.1.145	172.16.4.213	TCP	66	[TCP Retransmission] 55149 → 445 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170624	2481.9291116..	172.16.1.145	172.16.4.213	TCP	66	55154 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170630	2482.0304360..	172.16.1.145	172.16.4.213	TCP	66	55155 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170638	2482.2327429..	172.16.1.145	172.16.4.213	TCP	66	55156 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
170656	2482.6106411..	172.16.1.145	172.16.4.213	TCP	66	55157 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
171656	2581.5990662..	172.16.4.213	156.30.151.242	TCP	66	65301 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
172527	2562.1927544..	172.16.1.228	172.16.4.213	TCP	66	60490 → 5357 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
185209	2704.8073842..	172.16.4.213	52.98.20.130	TCP	66	65302 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
185404	2707.1187121..	172.16.4.213	40.126.62.32	TCP	66	65303 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
185405	2707.1194310..	172.16.4.213	40.126.62.32	TCP	66	65304 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
186505	2723.4204595..	172.16.4.213	104.18.26.44	TCP	66	65305 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
203483	2973.3172829..	172.16.4.213	150.171.27.12	TCP	66	65306 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
204478	2987.9014935..	172.16.4.213	140.82.112.26	TCP	66	65307 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM

> Frame 527: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface vde0
 > Ethernet II, Src: LiteonTechno_40:18:ed (e4:aa:ea:40:18:ed), Dst: RuijieNetwork_6a:15:26 (e0:54:00:00:00:00)
 > Internet Protocol Version 4, Src: 172.16.4.213, Dst: 23.196.227.9
 > Transmission Control Protocol, Src Port: 65322, Dst Port: 443, Seq: 0, Len: 0

0000 e0 5d 54 6a 15 26 e4 aa ea 40 18 ed 00 00 45 00 ..TJ&...@...E
 0010 00 34 fe f5 40 00 00 06 00 1c ac 10 04 d5 17 c4 ..4N@.....
 0020 e3 09 fe d0 01 bb 32 97 8c b0 00 00 00 00 80 022.....
 0030 fa f0 08 99 00 00 02 04 05 b4 01 03 03 08 01 01
 0040 04 02 ..

Around the 2476 second mark, a second internal host from a different subnet 172.16.1.x attempts a connection on port 80, followed by port 62078, port 445, and a massive burst to port 5357. This is suspicious as the execution is happening on the exact same port as the previous host 172.16.3.88, targeting the discovery port 5357 over and over again.

Capturing from Wi-Fi

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ip.addr == 172.16.3.88 && tcp.flags.syn == 1 && tcp.flags.ack == 1

No.	Time	Source	Destination	Protocol	Length	Info
125147	1763.1853129..	172.16.4.213	172.16.3.88	TCP	66	5357 → 65095 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125189	1763.2855929..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 50336 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125244	1763.5611196..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 64261 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125249	1763.6640106..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65322 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125254	1763.7270136..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65323 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125265	1763.8771556..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65324 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125282	1764.0104238..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65325 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125287	1764.1109005..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65095 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125369	1765.2941909..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 50336 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125385	1765.5748720..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 64261 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125388	1765.6755875..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65322 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125391	1765.7327389..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65323 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125405	1765.8920117..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65324 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125411	1766.0314110..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65325 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125417	1766.1250150..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 65095 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
125631	1769.3049525..	172.16.4.213	172.16.3.88	TCP	66	[TCP Retransmission] 5357 → 50336 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

> Frame 125053: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface vde0
 > Ethernet II, Src: LiteonTechno_40:18:ed (e4:aa:ea:40:18:ed), Dst: LiteonTechno_33:07:15 (b8:14:00:00:00:00)
 > Internet Protocol Version 4, Src: 172.16.4.213, Dst: 172.16.3.88
 > Transmission Control Protocol, Src Port: 5357, Dst Port: 50335, Seq: 0, Ack: 1, Len: 0

0000 b8 1e a4 33 07 15 e4 aa ea 40 18 ed 00 00 45 00 ...3...@...E
 0010 00 34 e5 90 40 00 00 06 b4 e5 ac 10 04 d5 ac 10 ..4..@.....
 0020 03 50 14 ed c4 9f 25 b4 40 f3 b3 a5 00 05 00 12 ..X...XW.....
 0030 ff ff 87 d3 00 00 02 04 05 b4 01 03 03 08 01 01
 0040 04 02 ..

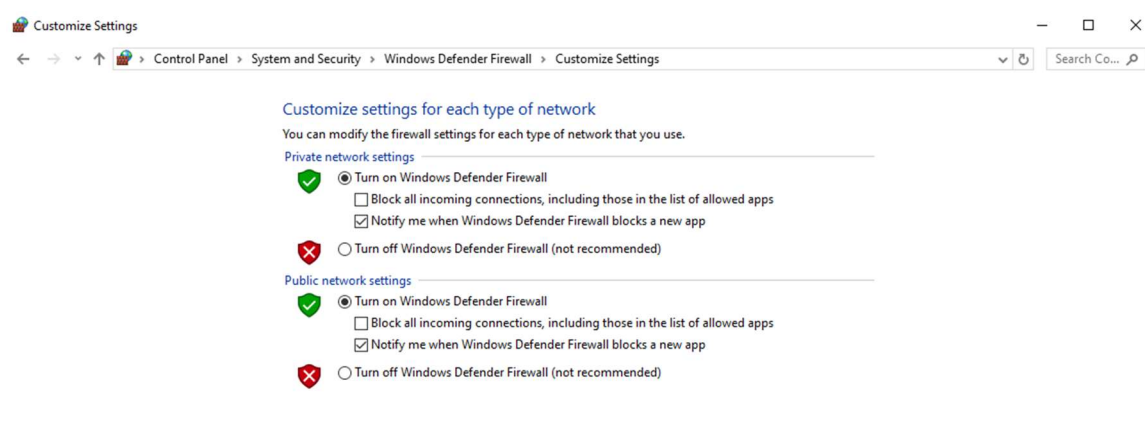
It raised much suspicion that port 5357 was being targeted so blatantly. The filter `ip.addr == 172.16.3.88 && tcp.flags.syn == 1 && tcp.flags.ack == 1` was used to filter for traffic between my host and the scanner, specifically looking for my host's replies. The results are as shown in the above picture. There is clear, unambiguous evidence of suspicious network activity.

When filtering for successful connection responses (`tcp.flags.syn == 1 && tcp.flags.ack == 1`), the capture reveals that the internal host 172.16.3.88 targeted my host (172.16.4.213) on Port 5357 (WS-Discovery).

My host replied with a SYN, ACK, confirming that Port 5357 is open and listening. However, the scanner immediately abandoned the connection without completing the

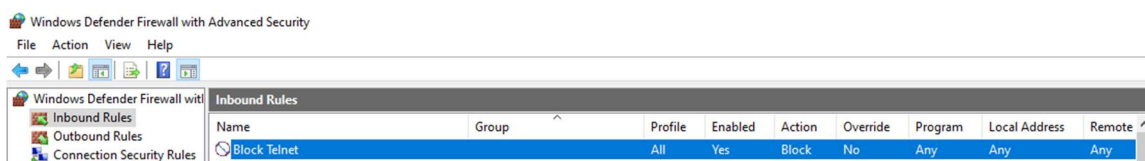
three-way handshake. This forced my host to generate a massive wave of dark-colored TCP Retransmissions trying to reach the scanner's ephemeral ports. This behavior is a textbook signature of an automated TCP SYN Stealth Scan, proving active local network reconnaissance.

Configuring Windows Defender Firewall



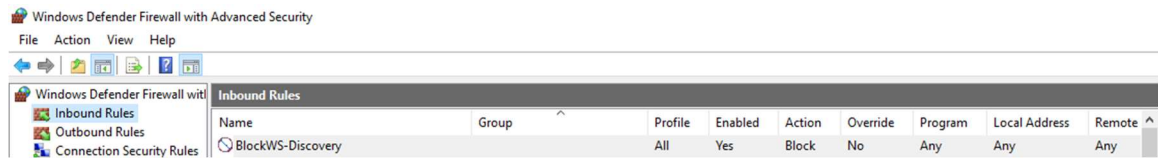
After recognizing the suspicious traffic on Wireshark, it was necessary to make sure that my network, be it public or private, was protected sealed. This step is important to filter traffic, block unauthorized access, protect systems from attacks, and restrict unsafe communication.

Block Telnet Rule (Proactive Hardening)



A firewall rule was created to block TCP port 23 used by Telnet. This step was especially necessary as Telnet is considered insecure as it transmits usernames and passwords in plain text without encryption. This demonstrated reducing vulnerabilities, restricting insecure services, and improving network protection.

Block WS Discovery (Evidence-Based Hardening)



By

blocking port 5357 (WS-Discovery), a demonstration of Incident Response and Threat Hunting was made. This rule represents an evidence-based security triage. An adversarial probe was observed, validated that the host was exposed and deployed a firewall rule to actively mitigate an ongoing threat vector.

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